

Literature Review

Critique

The strongest contribution is not simply that an LLM can extract epilepsy concepts, but that a transparent, auditable, modular pipeline can expose and control the trade-offs that matter in clinical use.

A safer and stronger claim is that you directly compare these approaches on the same epilepsy extraction tasks, with the same schema, the same evidence requirements, the same evaluation harness, and the same error taxonomy.

Evidence trails are central to the thesis, but reviewers will ask whether they are faithful or merely explanatory.

Modularity can sound like engineering unless the ablations show that each stage contributes measurable value. The full system does not need to win every aggregate metric. It needs to show a better clinical operating point: fewer unsupported outputs, clearer uncertainty handling, stronger temporal logic, and more predictable errors.

Configuration	What it tests	Signal to report
LLM-only	How far a prompted model gets without pipeline controls	Baseline semantic extraction and hallucination profile
Rules-only	How precise and brittle deterministic extraction is	High precision zones, recall gaps, edge-case failure modes
LLM + schema validation	Whether structural constraints reduce invalid outputs	Schema-validity gain and rejected-output rate
LLM + evidence verifier	Whether unsupported outputs fall	Unsupported extraction rate and evidence alignment
LLM + concept normalisation	Whether semantic equivalence improves scoring	Normalisation errors and per-concept F1
LLM + temporal resolver	Whether seizure-frequency logic improves	Temporal contradiction and current-frequency accuracy
Full hybrid system	Whether the architecture adds net value	Use-case-specific operating point, reliability, auditability

A more defensible phrase is 'LLM-assisted semantic interpretation and temporally aware clinical inference, constrained by deterministic schema validation and evidence verification.'

An interactive audit interface exposes intermediate representations, evidence links, final document annotations, and configuration-dependent behaviour.

Polished thesis statement

We present a modular, evidence-grounded hybrid LLM framework for clinical information extraction from epilepsy clinic letters. Unlike monolithic LLM or rules-based pipelines, the system decomposes extraction into configurable stages for semantic interpretation, relation extraction, temporal resolution, concept normalisation, schema validation, and evidence verification. We evaluate the framework across two independently developed, publicly shareable epilepsy benchmarks that stress different capabilities: broad phenotype extraction in ExECTv2 and temporally complex seizure-frequency extraction in Gan-style synthetic letters. Through direct comparison of rules-only, LLM-only, and hybrid configurations, plus component ablations and error analysis, we show how modular evidence-grounded design changes not only aggregate performance but also reliability, auditability, and use-case-specific trade-offs.

Your strongest empirical claim is cross-benchmark, cross-task robustness within epilepsy clinical letters. Your strongest architectural claim is that the same modular design can be configured for downstream use cases that prioritise recall, precision, uncertainty categorisation, or temporal timeline construction.

Evaluation checklist

Dimension	Report
Extraction	Per-class precision, recall, F1; micro and macro averages; calibration where relevant.
Evidence	Evidence span precision/recall, unsupported-output rate, wrong-evidence-right-label rate.

Dimension	Report
Schema	Validity rate, repair rate, rejected-output rate, invalid relation patterns.
Normalisation	Concept mapping accuracy, synonym handling, ontology or schema-level equivalence.
Temporality	Current-vs-historical status, frequency interval logic, contradiction resolution, seizure-free duration handling.
Uncertainty	Unknown frequency, no-seizure statements, ambiguous statements, and human-adjudication categories.
Robustness	Per-dataset performance, per-letter difficulty, bootstrap confidence intervals, and sensitivity to prompt/configuration changes.
Clinical utility	Operating points for trial recruitment, clinical care, timeline construction, and audit review.

Clinical Information Extraction Systems

Clinical information extraction (clinical IE) converts unstructured clinical narratives into structured entities, attributes, relations, codes and patient-level variables. In practice, IE is not merely named entity recognition. Robust clinical IE must also handle assertion status, temporality, experiencer, abbreviations, terminology mapping, evidence provenance, document-level aggregation and workflow governance. The literature shows a clear movement from symbolic rule systems to modular hybrid NLP pipelines, then to encoder language models and more recently to generative LLM systems. The most promising recent systems are not purely generative. They deliberately decompose the work into evidence selection, retrieval, field extraction, normalization, schema validation and human review.

Deterministic systems use dictionaries, grammars, ontologies, regular expressions, section rules and cue/scope logic. Classic systems such as MedLEE, MetaMap, ConText and MedEx are important because they demonstrate that explicit clinical semantics can be extracted reliably when the domain and output representation are well specified [2-4]. The defining feature of rules-based systems is explicit task decomposition. A note is sectioned, tokenized and searched for terms; context cues determine whether mentions are negated, historical, hypothetical or about someone other than the patient; then values are normalized to codes, units or controlled vocabularies. This makes the system auditable but can make it brittle across sites and documentation styles.

Hybrid systems are methodologically important because they make intermediate representations visible.

Clinical IE often requires several logically distinct decisions: finding an evidence span, determining assertion status, linking modifiers, resolving temporality, normalizing to a terminology and choosing the right patient-level value.

A modular extractor separates clinical IE into explicit stages. Older modular pipelines emphasized linguistic preprocessing and terminology mapping. Newer LLM-hybrid systems emphasize evidence retrieval, prompt/schema control, deterministic validation and human review. The purpose of modularity has shifted: from making text parsable to making model behavior governable.

Deterministic and statistical components select candidate evidence, context modules characterize that evidence, retrieval narrows the input, and validators check the output before it reaches downstream systems.

Retrieval reduces the amount of text sent to the model and makes the extraction step easier to audit.

Validators convert a fluent model output into a governed clinical data object.

Human review should not be a last-minute manual check of every field. It should be targeted by uncertainty, contradiction, low evidence quality, high clinical impact and drift.

Deterministic and hybrid systems expose the clinical meaning pipeline: entity, assertion, temporality, relation and normalization. Monolithic LLM systems compress this into a single prompt or model call. This simplification is useful during prototyping but weakens error analysis and audit.

Four decomposition strategies recur in the literature. Linguistic decomposition separates sectioning, tokenization, parsing and dictionary lookup. Clinical-context decomposition separates assertion, experiencer, temporality and certainty. Schema decomposition extracts one controlled field at a time. Retrieval decomposition first identifies relevant evidence, then asks a model to extract from that constrained context.

Future systems should combine deterministic and probabilistic strengths: rules for constraints, ontology mapping and validation; ML or LLM components for semantic flexibility; retrieval for evidence control; and human review for uncertainty and safety.

Every extracted value should be linked to evidence. The system should distinguish positive evidence, absent information, contradictory evidence and insufficient evidence. This enables safer abstention and more efficient human review.

Clinical LLMs

Clinical information extraction is shifting from isolated label prediction toward structured, provenance-bearing outputs that answer three linked questions: what was extracted, when it was true, and why the system believes it.

The strongest emerging designs are hybrid rather than purely generative. They combine constrained schemas, decomposed temporal tasks, retrieval or entity-centred context selection, span-level evidence, calibrated uncertainty estimation, abstention, and human review. Accuracy alone is no longer a sufficient evaluation target.

For clinical IE, the central problem is not merely whether a system can identify a disease, drug, biomarker, or event. A clinically useful extractor often needs to answer: when did it occur, how certain is the system, and what evidence in the record supports the extraction?

Clinical notes often mention a condition that is historical, negated, suspected, ruled out, copied forward, family history, planned, or superseded. Statements such as no evidence of recurrence, started chemotherapy in 2021, will hold pembrolizumab, and prior MI in 2016 are all clinically meaningful, but they should not be collapsed into the same present-tense positive label.

Later studies increasingly decomposed extraction into smaller steps. Wang et al. (2024) proposed a modular local LLM pipeline for pregnancy admission records, dividing extraction into concept extraction, aggregation, question generation, corpus extraction, and QA-style scale extraction. Pathology extraction studies point to the same lesson: models often recognise medical language, but fail when asked to resolve the correct temporal, anatomical, or evidential context.

LLMs can often identify temporal cues, but robust clinical temporal reasoning remains difficult.

Swaminathan et al. (2024) showed how clinical classifiers can abstain when uncertainty is high, improving clinical utility when the cost of a wrong prediction exceeds the cost of no prediction.

Gero et al. (2023) proposed self-verification: after producing an extraction, the LLM identifies provenance spans and checks its own outputs. Lopez et al. (2025) introduced CLEAR, which retrieves windows around clinical entities before LLM classification, improving grounding and reducing token cost. Windisch et al. (2026) represents a related trend: requiring verbatim evidence that can be mechanically validated.

Clinical IE should retrieve the smallest sufficient evidence context and organise longitudinal inputs into timelines, episodes, sections, or evidence packets.

JSON outputs, enumerated labels, binary temporal decisions, and explicit uncertain categories reduce ambiguity and support validation.

LLMs struggle most when time is implicit, relative, distributed across notes, or clinically encoded in treatment sequences.

Long context reduces truncation but can increase distraction.

Self-reported confidence should not be treated as calibrated confidence.

Evidence grounding improves auditability but does not eliminate temporal or inferential error.

report accuracy, calibration, evidence validity, abstention behaviour, and efficiency together.

Clinical LLM IE is converging on a clear lesson: accuracy alone is not enough. For high-value clinical extraction, the model must also know when an event happened, whether its answer is uncertain, and which text supports it.

Information Extraction from Clinical Notes and EHRs in Epilepsy and Related Neurological Conditions

The epilepsy clinical-note information extraction landscape is real, but not yet a large mature ecosystem. The most developed lines of work focus on seizure outcomes - especially seizure frequency, seizure freedom, and date of last seizure - plus broader clinic-letter phenotyping and epilepsy surgery referral-candidate identification. Across adjacent neurology domains, the same pattern recurs: clinically important outcomes are documented in narrative form rather than as clean structured variables. In multiple sclerosis this may be EDSS; in stroke it may be modified Rankin Scale, NIHSS, or functional status; in dementia it may be cognitive decline, biomarkers, family history, or functional impairment.

The largest unresolved issue is generalizability: documentation style varies enough across institutions that locally strong systems can degrade sharply when externally validated.

Penn / UPenn: seizure frequency and seizure freedom extraction

One influential line of work frames seizure outcome extraction as a machine-reading task: classify whether a patient is seizure-free, and extract text spans for seizure frequency and date of last seizure from clinic notes.

Xie et al. annotated 1,000 notes and fine-tuned BERT-family models, reporting near-human performance, with Bio_ClinicalBERT exceeding 80% accuracy for seizure-free classification and extraction F1 scores above 0.80. [1] The same group then applied models longitudinally to 55,630 notes from 9,510 patients at Penn between 2010 and 2022. Extracted outcomes allowed study of a relapsing-remitting disease course:

Decker et al. built a rule-based NLP algorithm to extract seizure types and frequencies from outpatient EHR notes. Internal performance was strong for a rule-based system, but external performance at a second epilepsy centre dropped sharply.

The ExECT work is especially relevant because it tries to build the kind of deep phenotype that structured EHR data rarely contains: semiology, syndrome, frequency, medication, EEG/MRI, comorbidity, and care history.

Holgate et al. used Llama 2 to extract seizure frequency from NHS outpatient epilepsy letters. The work highlights two durable challenges: seizure-frequency statements are often noisy or absent, and confidential EHRs must remain within secure hospital environments.

Gan et al. proposed evidence-grounded synthetic epilepsy letters for seizure-frequency inference.

A 2022 JMIR Medical Informatics paper extracted or predicted EDSS scores from 17,452 EHR records covering 4,906 MS patients. A combined keyword-CNN model performed best, with reported accuracy of 0.90 and F-score of 0.83.

Oh et al. developed an NLP pipeline to extract Alzheimer's-related phenotypes, including comorbidities, biomarkers, neurobehavioral scores, cognitive-decline indicators, family history, and neuroimaging findings. Reported phenotype-level F1 ranged broadly, reflecting different levels of extraction difficulty.

A 2024 systematic review of NLP for ADL extraction found 22 studies and 20 systems. It identified recurring barriers: restricted EHR access, small datasets, inadequate documentation, limited granularity, and heterogeneous outcome definitions.

Recurring challenges

- Inconsistent documentation: Seizure status may be stated as exact counts, vague categories, relative change, or absence of events. Copy-paste and contradictions are common in clinical notes.
- Multiple event types: A single epilepsy note may distinguish tonic-clonic seizures, focal aware seizures, auras, absences, myoclonus, and non-epileptic events. One frequency value is often insufficient.
- Temporality: Models must separate current frequency, historical frequency, last seizure date, seizure freedom since last review, and lifetime history.
- External generalizability: Performance can degrade when note style changes across centres, specialties, templates, and documentation cultures.

- Gold-standard annotation cost: Clinician annotation is expensive and ambiguous; even experts disagree on onset, frequency, temporality, and certainty.

Epilepsy NLP and LLM Literature Review

Across the papers, the shared problem is that clinically decisive epilepsy information is present in narrative text, but not reliably available as computable data. Seizure frequency, seizure freedom, seizure type, epilepsy type, medication history, and temporal relations are central to care, safety counselling, research cohorts, quality metrics, and risk stratification. Yet these variables are often embedded in outpatient letters, hospital reports, EMU reports, and clinical notes rather than in structured fields. This creates a gap between what clinicians document and what researchers or health systems can measure at scale.

The stakes are therefore both clinical and infrastructural. Clinically, mischaracterising seizure burden can affect medication changes, surgical referral, safety advice, and evaluation of seizure control. Infrastructurally, the lack of structured seizure data limits registries, audit, longitudinal outcome studies, and multicentre research.

These studies reposition the challenge from "can we build rules or train a classifier?" to "can we reliably convert messy clinical prose into structured, auditable, temporally precise data with less annotation burden?"[3][4][11][12]

Seizure burden is not just a phrase; it requires anchoring to time, dates, note context, and sometimes seizure type.

Annotation burden and privacy constraints shape what can be validated and reused.

A system can have a high F1 score on one corpus but still be weak if it ignores seizure type, cannot resolve dates, hallucinates structured fields, requires a closed API, or fails outside specialist epilepsy notes.

LLM pipeline papers such as Dao et al., Wang et al., and CLINES place more emphasis on structured-output reliability than on token-level preprocessing. Their key post-processing elements include validation, retry logic, schema enforcement, unit normalisation, UMLS normalisation, and date resolution. These are highly relevant to epilepsy because seizure outcomes must be converted into computable values rather than left as free-text summaries.

Seizure outcomes are temporal: "seizure-free," "last seizure," "weekly," "since medication change," "before surgery," and "during childhood" all require temporal anchoring.

Future systems should represent seizure events, frequencies, time windows, medication changes, and clinical encounters in a structured temporal model rather than as isolated spans.

Generative LLMs introduce risks that BERT-style span extractors do not: hallucinated fields, inconsistent formats, over-explanation, and unsupported inferences. Holgate et al. note hallucinations and false positives as continuing issues; Dao et al. respond to this class of problem with guardrails, validation, error detection, and retry logic. Gan et al.'s evidence-span approach is also important because it lets a user check whether a structured output is grounded in the source text. Future epilepsy LLMs should output not only values but also evidence spans, confidence estimates, normalised units, temporal anchors, and provenance.

The central clinical motivation remains stable: epilepsy care depends on seizure outcomes that are often trapped in prose. The technical challenge has evolved: the problem is no longer just finding seizure terms, but producing temporally precise, normalised, evidence-grounded, institutionally robust, and auditable structured data.

Hybrid Deterministic + LLM Extraction of Structured Seizure Phenotypes

The strongest publishable framing is not simply that a large language model can read epilepsy clinic letters. It is that a clinically grounded, evidence-preserving, schema-constrained hybrid pipeline can extract seizure phenotypes from narrative letters more transparently than an LLM-only system and more flexibly than a deterministic-only system.

how can epilepsy NLP research become reproducible, clinically meaningful, and shareable when the data are narrative, private, heterogeneous, and full of temporal ambiguity?

Do not ask only, "Which model has the highest F1?" Ask: "What kind of clinical truth is the system trying to extract? What evidence is the output grounded in? What does the schema make visible, and what does it suppress? What kinds of mistakes would be clinically tolerable, and what kinds would be unacceptable?" Epilepsy clinic letters are not merely administrative correspondence. They are compact clinical narratives that often contain the most important longitudinal facts about a patient's epilepsy: diagnosis, epilepsy type, seizure type, semiology, frequency, date of last seizure, medication history, investigations, side effects, comorbidities, and plans. Much of this material is not reliably stored in structured fields, and even when structured fields exist they may lag behind the nuanced account contained in free text. This is the central motivation behind ExECT and the broader epilepsy NLP literature (Fonferko-Shadrach, Lacey, White, et al. 2019; Yew et al. 2023; Xie et al. 2023b).

The clinical difficulty is that a seizure phenotype is not a single label. A useful phenotype is a temporally anchored, assertion-qualified, clinically interpreted bundle of information. A letter may say that the patient had focal impaired awareness seizures in childhood, now has rare nocturnal convulsions, has weekly auras, is seizure-free from disabling seizures, and remains on lamotrigine while levetiracetam was stopped because of mood effects. A flat extractor that returns only "focal seizure" or "monthly" would miss much of the clinical meaning.

You are extracting the phenotype as documented in the letter, ideally with evidence spans, temporality, assertion status, and uncertainty. That distinction is not pedantic. It is what separates a responsible clinical NLP system from an overconfident summariser.

Every clinically important extraction should have at least five attributes: a controlled label, the original wording or evidence span, assertion status, temporality, and uncertainty. For frequency and dates, add a normalised value and an anchor date. Without these attributes, performance metrics can look good while the output remains clinically brittle.

The 2019 ExECT paper developed a system in GATE that combined rule-based and statistical methods to extract structured epilepsy data from clinic letters, including diagnosis, epilepsy type, seizure type, seizure frequency, anti-seizure medication, and investigations (Fonferko-Shadrach, Lacey, White, et al. 2019). Its reported performance was clinically promising but uneven across fields: medication extraction was strong, while seizure frequency and some seizure-type categories had lower recall.

That unevenness is not a flaw to dismiss. It is a clue. Structured medication statements are often written in repeatable forms: drug name, dose, unit, frequency, action. Seizure frequency is more narrative. Some parts of the letter behave like forms; others behave like stories.

Fonferko-Shadrach, Lacey, Akbari, et al. (2024) is easy to underestimate if it is read merely as "the paper that releases synthetic annotated letters." Its deeper contribution is epistemic. It asks what it means to create a clinically credible gold standard for epilepsy NLP when real letters cannot be openly shared and when even trained annotators disagree.

The reported inter-annotator agreement is revealing. Overall agreement was moderate rather than perfect, with some entities much easier than others. Prescriptions were comparatively reliable; "when diagnosed" and other temporally or narratively embedded concepts were harder. ExECTv2 achieved high overall scores against the consensus standard, but seizure frequency again remained one of the weaker categories (Fonferko-Shadrach, Lacey, Akbari, et al. 2024). The message is not simply that NLP is hard. It is that the gold standard is itself a clinical-linguistic artefact, shaped by guideline clarity, concept boundaries, and adjudication decisions.

When your system disagrees with the benchmark, is the system wrong, the annotation under-specified, or the clinical sentence genuinely ambiguous? A sophisticated dissertation should not treat the gold standard as infallible; it should treat it as the best available operationalisation of a difficult clinical reading task.

A deterministic-only baseline can be credible because ExECT itself is rooted in deterministic and rule-based clinical NLP. An LLM-only baseline can test whether modern models can subsume some of that engineering. The hybrid system can then argue for complementarity: deterministic components retrieve candidate evidence, normalise outputs, enforce schema validity, and flag contradictions, while the LLM resolves semantic ambiguity and generates structured interpretations from messy text.

Seizure frequency looks like a numeric extraction problem until one reads real letters. Then it becomes obvious that it is a temporal reasoning problem, an event-typing problem, and sometimes a clinical uncer-

tainty problem. The literature repeatedly returns to seizure frequency because it is both clinically central and linguistically awkward.

Frequency can be explicit: "two seizures per month." It can be approximate: "about once every few weeks." It can be interval-based: "three since January." It can be negative: "no seizures since surgery." It can be event-type specific: "auras weekly, convulsions once a year." It can be embedded in clusters: "two clusters of three to four seizures." It can be unknown: "not clear how often these occur." It can also be absent, which is not the same as unknown. These distinctions are central in Gan, Roy, et al. (2026), but they are foreshadowed throughout the earlier literature.

A dissertation that compares local open models with API-based LLMs could frame the comparison not only around accuracy but around deployability, privacy, reproducibility, and cost.

ExECT is broad, public, clinically authored, and annotation-guideline driven. Gan is narrow, synthetic-supervision driven, evidence-aware, and focused on the difficult normalisation problem of seizure frequency. One tests breadth of epilepsy phenotyping; the other tests depth of a high-value temporal phenotype.

Deterministic methods are precise, auditable, and excellent for lexical regularities, but brittle under paraphrase and institutional variation. Transformer and LLM methods are more flexible, but they can over-infer, hallucinate unsupported answers, mishandle temporality, and obscure the reason for an output.

The hybrid argument is strongest when it is operational rather than rhetorical. You should be able to point to a specific error and say which component should prevent it. For example, if an LLM extracts "monthly seizures" from a sentence about childhood seizures, the deterministic temporal filter or assertion validator should flag the evidence as historical. If a rule system misses "episodes of blankness and lip-smacking," the LLM should map it to a candidate seizure description while preserving the original wording and uncertainty. If a frequency answer has no evidence span, the validator should reject it or mark it for human review.

A useful hybrid architecture starts by refusing to use the LLM for tasks that do not need it. This reduces cost, improves reproducibility, and makes the system easier to explain.

The LLM should be used where the letter requires semantic interpretation: linking a frequency to the correct seizure type, distinguishing current from historical narrative when the wording is subtle, interpreting semiology, handling uncertain or hedged statements, and resolving local contradictions into a structured answer with evidence. It should not be allowed to silently invent labels. It should be asked to cite evidence and express uncertainty.

Suggested high level pipeline:

1. Deterministic preprocessing: split sections, anchor dates, identify medication and seizure-term candidates.
2. Evidence retrieval: collect sentences and neighbouring context likely to contain phenotype evidence.
3. Schema-constrained LLM extraction: output JSON with labels, evidence spans, temporality, assertion, uncertainty, and normalised values.
4. Deterministic validation: check JSON validity, allowed ILAE labels, evidence support, date/frequency consistency, medication status, and contradictions.
5. Review flags: mark unsupported, contradictory, uncertain, legacy-term, no-evidence, or low-confidence outputs for human review.

This design resonates with general clinical IE work outside epilepsy. Liu et al. (2025) use rule-based filtering and LLM classification for statin information extraction, illustrating how deterministic components can reduce irrelevant text while preserving recall. Dao et al. (2025) describe a schema-oriented generative AI extraction pipeline with validation and guardrails. These papers are useful not because statins or general medical extraction are the same as epilepsy, but because they support the idea that production-grade clinical LLM systems need scaffolding.

Evidence-span requirements counter this tendency. They force the system to answer: which words in the letter justify the extraction? They also make error analysis more informative. An unsupported answer, a wrong interpretation of correct evidence, and a correct interpretation of wrong evidence are different failures. Without evidence, they all look like the same binary error.

In a short IEEE paper, annotation often gets compressed into a paragraph. For your dissertation thinking, it should be central. The quality of the system is bounded by the quality and clarity of the target labels.

The relevant question is not just annotation agreement but label grammar: what does each frequency category mean, how are ranges mapped, how is seizure-free duration treated, and how are "unknown" and "no seizure" separated? If you have direct access to the synthetic letters, read the label scheme before reading the model results. The schema is where the clinical assumptions live.

A strong evaluation should compare deterministic-only, LLM-only, and hybrid systems. Without those baselines, the word "hybrid" will sound like architecture preference rather than evidence. The ablation should show not only whether the hybrid improves headline F1 but why it improves clinically relevant behaviour. You should also measure properties that classic NLP metrics miss. Unsupported-output rate: how often does the system make a claim with no evidence? Invalid-output rate: how often does the LLM produce malformed JSON or illegal labels? Contradiction rate: how often does the output conflict with another extracted fact? Human-review burden: what proportion of letters are flagged, and are the flags useful?

Generalisation is one of the most important themes in this literature. Decker et al. (2022) show rule-based performance can drop sharply externally. Xie et al. (2023a) examine how fine-tuned transformer models behave in new clinical contexts.

A good error analysis can make a modest model result interesting. In clinical NLP, reviewers often trust a system more when the authors understand its failures. For seizure phenotype extraction, the error taxonomy should be clinically informed, not just technical.

Failure mode	Typical example	Hybrid mitigation
Historical vs current seizures	Childhood tonic-clonic seizures ex- tracted as current events.	Section and temporal cue rules; LLM temporality classification; validator requiring current evi- dence.
No seizure vs unknown	Silence about seizures treated as seizure freedom, or vague text treated as no seizure.	Require explicit evidence for no- seizure labels; allow unknown as valid output.
Multiple event types	Auras weekly, convulsions yearly, absences absent.	Event-type linking; separate frequency objects per seizure type.
Family history	"Mother has epilepsy" extracted as patient diagnosis.	Family-history section/kin- ship cue rules; assertion- status check.
Functional or non- epileptic attacks	PNES or dissociative seizures mapped to epileptic seizure types.	Controlled labels for non- epileptic events; uncertainty flags; clinician-review category.
Medication status	Stopped or planned medication ex- tracted as current.	Medication-action lexicon; cur- rent/past/planned status classi- fier.
Legacy terminology	"Complex partial" not mapped, or mapped without preserving original text.	ILAE synonym map plus original-surface-text field.
Relative dates	"Six weeks ago" not anchored to letter date.	Date anchor extraction; relative- date normaliser; uncertainty if anchor missing.
Contradictory state- ments	"No seizures since 2022" and "possi- ble nocturnal event last month."	Contradiction detector; output multiple hypotheses or review flag.

The dissertation can be written as a story about complementarity. The opening problem is that epilepsy phe- notypes are clinically important but trapped in unstructured letters. The related work shows a progression: rule-based systems such as ExECT prove the value of epilepsy-specific NLP; transformer machine-reading approaches improve seizure outcome extraction; LLMs introduce flexible semantic extraction but raise con- cerns about unsupported inference; synthetic data offers reproducibility and privacy but needs real-world anchoring. The gap is not "no one has used AI for this." The gap is how to combine deterministic clinical constraints, LLM semantic flexibility, and evidence-grounded evaluation for auditable seizure phenotype extraction.

The paper could be organised around four claims:

1. Clinical claim: seizure phenotypes in clinic letters require temporality, assertion, evidence, and schema grounding, not just entity detection.
2. Methodological claim: deterministic components and LLMs solve different subproblems and should be combined rather than treated as alternatives.

3. Benchmark claim: ExECT and Gan provide complementary reproducible benchmarks for broad and narrow phenotype extraction.
4. Evaluation claim: the system should be judged not only by F1 but also by unsupported-output rate, schema validity, temporality errors, and clinical error severity.

The paper will feel publishable if it does four things well.

- First, it must define the extraction target precisely. "Seizure phenotype" should not float as a vague phrase. Specify which fields are extracted, how they map to ILAE terminology, how temporality and assertion are represented, and what counts as evidence.
- Second, it must include meaningful baselines. Deterministic-only and LLM-only systems are not optional; they are necessary to justify the hybrid. Add ablations for evidence retrieval, schema validation, and post-processing if time permits.
- Third, it must report per-field and error-type results. Overall F1 is useful but insufficient. The reader should know whether the system succeeds because it extracts medications well or because it truly improves difficult seizure-frequency and temporality fields.
- Fourth, it must be honest about generalisation. ExECT and Gan are excellent benchmarks, but they are not proof of safe deployment. A clear limitations section that distinguishes benchmark performance from prospective clinical utility will make the work more credible, not less.

The literature points toward a simple but demanding conclusion: epilepsy phenotype extraction is not just a natural language problem. It is a clinical representation problem. The system must read text, but it must also respect the structure of epilepsy knowledge, the uncertainty of clinical documentation, the temporality of seizure histories, and the privacy constraints of health records.

Together, they allow a dissertation to ask a nuanced question: can a hybrid deterministic plus LLM pipeline preserve the auditability and schema discipline of traditional clinical NLP while gaining the interpretive flexibility of modern language models?

That question is worth pursuing because it is not a fashion statement about LLMs. It is a practical question about how clinical AI systems should be built when the information is important, messy, private, and only partly reducible to rules.

Uncertainty Estimation, Evidence Verification and LLM Reliability

Wong et al., however, found persistent overconfidence and weak failure prediction, especially for difficult tasks [5]

RAG evaluation systems such as RAGTruth, RAGAS, and ARES then diagnose retrieval and generation failures in source-grounded systems [15-17]

The most recent trend is integration. Self-RAG trains models to retrieve, generate, and critique with reflection tokens [18]. FacTool, Chain-of-Verification, RARR, ProgramFC, and FIRE treat verification as a multi-step process involving search, decomposition, judging, revision, and sometimes stopping decisions based on confidence [19-23]. Conformal, refusal-aware, and calibration-tuning approaches attempt to turn uncertainty estimates into guarantees, refusals, or better-calibrated responses [24-27].

Theme	Central question	Core signal	Best suited to	Main risk
1. Sampling and semantic uncertainty	Do multiple generations agree in meaning?	Self-consistency, semantic entropy	Open-ended QA, hallucination screening	Systematic wrong answers can appear consistent
2. Attribution and citation	Do cited sources support claims?	Citation support, AIS labels	Search, summarisation, answer engines	Citations create false reassurance if unchecked
3. RAG evaluation	Where did the RAG pipeline fail?	Context relevance, answer faithfulness, answer relevance	Enterprise RAG, grounded QA	Retriever and judge errors can mask failures
4. Agentic verification	Can the system actively check and revise itself?	Tool use, query planning, verification questions	Complex claims and long outputs	Cost, latency, circular LLM judging
5. Selective prediction and abstention	When should the system not answer?	Conformal sets, calibrated thresholds, refusal training	High-stakes or uncertain contexts	Abstention may reduce usefulness

Sampling-based approaches assume that knowledge produces stable outputs while uncertainty produces divergent outputs. SelfCheckGPT operationalises this idea by comparing multiple sampled passages from the same black-box model and identifying inconsistent sentences [6]. Semantic entropy takes a more

principled step by clustering sampled generations into semantic equivalence classes and estimating entropy over meanings rather than word forms [7]. Evaluation moves from 'is this answer right?' to 'which claims are supported, unsupported, contradicted, ambiguous, or unverifiable?' The cost is pipeline fragility. Decomposition, retrieval, entailment, and judging can all introduce errors.

RAG reduces reliance on parametric memory by providing retrieved context, but it does not guarantee faithfulness.

The strongest contribution of this theme is diagnostic separation. A RAG system can fail because retrieval missed evidence, retrieval returned noisy evidence, generation ignored evidence, or the final answer added unsupported claims. Evaluation therefore needs separate metrics for retriever quality, evidence sufficiency, answer faithfulness, and citation support.

Agentic systems treat verification as a workflow. Self-RAG trains a model to adaptively retrieve and critique [18]. FacTool uses task-specific tools across knowledge QA, code, math, and literature review [19]. Chain-of-Verification drafts an answer, plans verification questions, answers them independently, and revises [20]. RARR retrieves evidence after generation and edits unsupported content [21]. ProgramFC decomposes complex claims into program-guided sub-tasks [22]. FIRE iteratively retrieves and verifies until a confidence-based stopping rule is met [23].

This is the most deployment-like theme because real verification requires planning, tool use, and revision. It is also the most expensive. The main risk is circularity: LLMs may generate claims, search queries, evidence interpretations, and final judgments. Robust systems need audit sets, human spot checks, and careful separation between generation and judging.

This literature matters because hallucination is partly an incentive problem. When evaluations reward correct guesses and penalise abstentions, models are encouraged to answer even when uncertain. Nature's 2026 paper on accuracy-based evaluations argues that such incentives can reward unwarranted guessing [33].

Rigorous evaluation requires separating estimation quality, verification quality, and decision quality.

Evaluation target	Question answered	Representative metrics	Failure if omitted
Failure prediction	Can the method identify likely wrong answers?	AUROC, AUPRC, error-detection precision/recall	A well-calibrated average score can still fail on individual high-risk cases.
Selective decision	Does abstaining or retrieving reduce risk at acceptable coverage?	Selective risk, risk-coverage curves, abstention/error trade-off	A score may be informative but operationally useless.
Claim support	Which claims are supported, contradicted, or unverifiable?	Atomic support rate, factual F1, entailment labels, human agreement	Answer-level labels hide mixed truth values.
RAG diagnostics	Did retrieval, grounding, or generation fail?	Context relevance, answer faithfulness, answer relevance, span hallucination labels	Final answer accuracy cannot localise pipeline failures.

Automated LLM judges are useful but require meta-evaluation. They should be tested for sensitivity to wording, order, verbosity, and source presentation. When possible, automated scores should be calibrated against human labels, reported with uncertainty intervals, and stress-tested on adversarial examples.

Many detectors assume wrong answers are uncertain. This fails when a model gives a stable but false answer.

Evidence verification depends on source coverage and retrieval quality.

Systems that use LLMs to generate, retrieve, judge, and revise can have correlated errors. Human spot checks and independent audits remain important.

Accuracy-only scoring can penalise abstention and reward lucky guesses. Evaluation design should distinguish correct answers, incorrect answers, and justified non-answers

A confidence score alone is useful but insufficient. A self-consistency score is stronger, but can miss systematic hallucinations. A claim-level evidence check is more interpretable, but depends on retrieval and judging quality. A calibrated decision policy - answer, retrieve, revise, abstain, or defer - is the most deployment-relevant form of uncertainty estimation.

The strongest future systems will likely be hybrid: semantic uncertainty for early warning, internal-state signals for cheap monitoring, retrieval and citation checks for factual grounding, calibrated abstention for risk control, and audited LLM or human judges for high-stakes verification.